TP7 : Docker avancé et Docker Compose (multi-stage, non-root, PostgreSQL)

Environnement du TP

```
arij@vagrant:~$ docker version
Client:
 Version:           29.1.3
 API version:       1.52
 Go version:        go1.24.4
 Git commit:        29.1.3-0ubuntu3~24.04.2
 Built:             Wed Apr 29 16:41:06 2026
 OS/Arch:           linux/amd64
 Context:           default

Server:
 Engine:
  Version:          29.1.3
  API version:      1.52 (minimum version 1.44)
  Go version:       go1.24.4
  Git commit:       29.1.3-0ubuntu3~24.04.2
  Built:            Wed Apr 29 16:41:06 2026
  OS/Arch:          linux/amd64
  Experimental:     false
 containerd:
  Version:          2.2.1
  GitCommit:
 runc:
  Version:          1.3.4-0ubuntu1~24.04.1
  GitCommit:
 docker-init:
  Version:          0.19.0
  GitCommit:
arij@vagrant:~$ docker compose version
Docker Compose version 2.40.3+ds1-0ubuntu1~24.04.1
arij@vagrant:~$ docker buildx version
github.com/docker/buildx 0.30.1 0.30.1-0ubuntu1~24.04.1
```

Partie 1 : Couches et cache

1. télécharger les dépendances et créer le fichier package-lock.json

```
arij@vagrant:~/app-starter/app-starter$ npm install

added 92 packages, and audited 93 packages in 12s

29 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities
```

2. Construire l'image naïve

```
arij@vagrant:~/app-starter/app-starter$ time docker build -f Dockerfile.naive -t visits-app:naive .
[+] Building 142.7s (10/10) FINISHED                                                                                docker:default
 => [internal] load build definition from Dockerfile.naive                                                                   0.0s
 => => transferring dockerfile: 393B                                                                                         0.0s
 => [internal] load metadata for docker.io/library/node:24                                                                   2.7s
 => [internal] load .dockerignore                                                                                            0.0s
 => => transferring context: 2B                                                                                              0.0s
 => [1/5] FROM docker.io/library/node:24@sha256:d862bb5d082c5ae0210ac07e8c0344cb8cf7bef6d78e9b8c50c179da923108ea           124.9s
 => => resolve docker.io/library/node:24@sha256:d862bb5d082c5ae0210ac07e8c0344cb8cf7bef6d78e9b8c50c179da923108ea             0.0s
 => => sha256:2d5c7653879635dfef69f3b864d5a53e937507b1a6d4b3634f8636f2dd0d0626 448B / 448B                                   0.3s
 => => sha256:17110b031ffba487d7c86f79f9e6149f648fb23fe89036d660065e5afe5608dc 1.25MB / 1.25MB                               1.2s
 => => sha256:32321da9b5bae16da3ff55f9262b606c15abd172349c40babd6b0d78bfc59b70 60.04MB / 60.04MB                            77.7s
 => => sha256:4c32d53bc5f6ba8d133118f0184b96b44addea0bc86986f399cc859a2d0a7e51 3.33kB / 3.33kB                              1.0s
 => => sha256:c28c2f6c340d2f618d807e99ea6dd05b7cdc0cb80f44a32ef2511370286752fe 211.82MB / 211.82MB                          97.5s
 => => sha256:661ef7e1e7b1a9919ab256de1deb5be9107b28a15fea74b39921bec95b713038 64.43MB / 64.43MB                           35.4s
 => => sha256:49dd0ae7831394e9635b737d684810716dadcb662b5b9b80eefd89ceeld65395 23.87MB / 23.87MB                           35.9s
 => => sha256:a4aa977c0d0eb2336dad68941e458308ebf32a717c19ea72e475b9eda8035506 48.52MB / 48.52MB                           38.5s
 => => extracting sha256:a4aa977c0d0eb2336dad68941e458308ebf32a717c19ea72e475b9eda8035506                                    5.6s
 => => extracting sha256:49dd0ae7831394e9635b737d684810716dadcb662b5b9b80eefd89ceeld65395                                    1.1s
 => => extracting sha256:661ef7e1e7b1a9919ab256de1deb5be9107b28a15fea74b39921bec95b713038                                    7.0s
 => => extracting sha256:c28c2f6c340d2f618d807e99ea6dd05b7cdc0cb80f44a32ef2511370286752fe                                   19.6s
 => => extracting sha256:4c32d53bc5f6ba8d133118f0184b96b44addea0bc86986f399cc859a2d0a7e51                                    0.0s
 => => extracting sha256:32321da9b5bae16da3ff55f9262b606c15abd172349c40babd6b0d78bfc59b70                                    7.0s
 => => extracting sha256:17110b031ffba487d7c86f79f9e6149f648fb23fe89036d660065e5afe5608dc                                    0.1s
 => => extracting sha256:2d5c7653879635dfef69f3b864d5a53e937507b1a6d4b3634f8636f2dd0d0626                                    0.0s
 => [internal] load build context                                                                                            1.0s
 => => transferring context: 29.39MB                                                                                         0.9s
 => [2/5] WORKDIR /app                                                                                                       1.3s
 => [3/5] COPY . .                                                                                                           0.8s
 => [4/5] RUN npm install                                                                                                    2.9s
 => [5/5] RUN npm run build                                                                                                  6.1s
 => exporting to image                                                                                                       3.6s

real    2m22,929s
user    0m2,117s
sys     0m1,956s
arij@vagrant:~/app-starter/app-starter$
```

3. Modifier le code pour tester le cache

```
  GNU nano 7.2                                          src/server.ts *
import express from "express";
import { Pool } from "pg";

const pool = new Pool();
const app = express();
const port = Number(process.env.PORT ?? 3000);

async function init(): Promise<void> {
  await pool.query(
    "CREATE TABLE IF NOT EXISTS visits (id SERIAL PRIMARY KEY, at TIMESTAMPTZ DEFAULT now())"
  );
}

app.get("/", async (_req, res) => {
  await pool.query("INSERT INTO visits DEFAULT VALUES");
  const { rows } = await pool.query("SELECT count(*)::int AS total FROM visits");
  res.json({
    message: "Bonjour depuis le TP7 - Version modifiée pour le cache",
    visits: rows[0].total,
    host: process.env.HOSTNAME
  });
});

app.get("/health", async (_req, res) => {
  try {
    await pool.query("SELECT 1");
    res.json({ status: "ok" });
  } catch {
    res.status(503).json({ status: "db-unavailable" });
```

4. relancer le build

```
arij@vagrant:~/app-starter/app-starter$ time docker build -f Dockerfile.naive -t visits-app:naive .
[+] Building 10.0s (10/10) FINISHED                                                                  docker:default
 => [internal] load build definition from Dockerfile.naive                                                     0.0s
 => => transferring dockerfile: 393B                                                                           0.0s
 => [internal] load metadata for docker.io/library/node:24                                                     0.7s
 => [internal] load .dockerignore                                                                              0.0s
 => => transferring context: 2B                                                                                0.0s
 => [1/5] FROM docker.io/library/node:24@sha256:d862bb5d082c5ae0210ac07e8c0344cb8cf7bef6d78e9b8c50c179da923108ea  0.0s
 => => resolve docker.io/library/node:24@sha256:d862bb5d082c5ae0210ac07e8c0344cb8cf7bef6d78e9b8c50c179da923108ea  0.0s
 => [internal] load build context                                                                              0.1s
 => => transferring context: 71.29kB                                                                           0.1s
 => CACHED [2/5] WORKDIR /app                                                                                  0.0s
 => [3/5] COPY . .                                                                                             0.3s
 => [4/5] RUN npm install                                                                                      1.4s
 => [5/5] RUN npm run build                                                                                    2.5s
 => exporting to image                                                                                         4.6s
 => => exporting layers                                                                                        2.7s
 => => exporting manifest sha256:7958f09a1871429bab6a3ee33c0b50078bcfbc8f193a9e572e891c49c5d608fb              0.0s
 => => exporting config sha256:14d6034829412a2f0dfb5b6bb5065449edcdbe2d5d4fc18258c60193ffe8b74d                0.0s
 => => exporting attestation manifest sha256:9b36c1e0f06f21edba2944d99b66ce85fa1cdd2e72875e70cda59c833b3335ea  0.1s
 => => exporting manifest list sha256:55ef1630e69ed119b477fd788c90ffb15933e281620253e6f61b94b683106c0f         0.0s
 => => naming to docker.io/library/visits-app:naive                                                            0.0s
 => => unpacking to docker.io/library/visits-app:naive                                                         1.6s

real    0m10,188s
user    0m0,188s
sys     0m0,181s
```

5. Afficher les couches et leur taille :

```
arij@vagrant:~/app-starter/app-starter$ docker image ls visits-app
                                                                                    i Info →  U In Use
IMAGE              ID             DISK USAGE   CONTENT SIZE   EXTRA
visits-app:naive   55ef1630e69e       1.67GB          417MB
arij@vagrant:~/app-starter/app-starter$ docker run --rm visits-app:naive id
uid=0(root) gid=0(root) groups=0(root)
```

Partie 2 : Mesurer l'image naïve

1. Lancer le scan Trivy sur l'image naïve

```
arij@vagrant:~/app-starter/app-starter$ docker run --rm -v /var/run/docker.sock:/var/run/docker.sock \
  aquasec/trivy:latest image --severity HIGH,CRITICAL --quiet visits-app:naive
Unable to find image 'aquasec/trivy:latest' locally
latest: Pulling from aquasec/trivy
44136fa355b3: Download complete
6b85eccf6d37: Pull complete
7982da00fb0e: Pull complete
2e51b4bff2eb: Pull complete
e2de96513ba9: Pull complete
2a4d4a16b7c4: Download complete
f6da3aa7a7da: Download complete
Digest: sha256:790fd66d1697395d8693d1acaa1ea702406a20f109b6fc7bac8db623eca6c06c
Status: Downloaded newer image for aquasec/trivy:latest

Report Summary

┌──────────────────────────────────────────────────────────────────────────────┬────────┬─────────────────┬─────────┐
│                                    Target                                    │  Type  │ Vulnerabilities │ Secrets │
├──────────────────────────────────────────────────────────────────────────────┼────────┼─────────────────┼─────────┤
│ visits-app:naive (debian 12.15)                                              │ debian │       479       │    -    │
├──────────────────────────────────────────────────────────────────────────────┼────────┼─────────────────┼─────────┤
│ app/node_modules/@types/body-parser/package.json                             │ node-pkg │       0       │    -    │
├──────────────────────────────────────────────────────────────────────────────┼────────┼─────────────────┼─────────┤
│ app/node_modules/@types/connect/package.json                                 │ node-pkg │       0       │    -    │
```

```
┌──────────────────────────────────────┬──────────────────┬──────────┬──────────┬──────────────────────────────┬──────
│                                      │ CVE-2026-109869  │          │          │                              │ un
dici: undici: Denial of Service via unhandled stream error │
│                                      │                  │          │          │                              │ on
 excessive content...                  │
│                                      │                  │          │          │                              │ ht
tps://avd.aquasec.com/nvd/cve-2026-109869
│                                      ├──────────────────┤          │          ├──────────────────────────────┼──────
│                                      │ CVE-2026-19534   │          │          │ 6.28.1, 7.29.1, 8.10.2       │ un
dici: undici: Denial of Service via unrequested WebSocket  │
│                                      │                  │          │          │                              │ su
bprotocol                             │
│                                      │                  │          │          │                              │ ht
tps://avd.aquasec.com/nvd/cve-2026-19534
│                                      ├──────────────────┤          │          ├──────────────────────────────┼──────
│                                      │ CVE-2026-87018   │          │          │ 6.29.1, 7.30.1, 8.12.2       │ un
dici: undici: Denial of Service via TLS renegotiation with │
│                                      │                  │          │          │                              │ mi
smatched certificate                  │
│                                      │                  │          │          │                              │ ht
tps://avd.aquasec.com/nvd/cve-2026-87018
└──────────────────────────────────────┴──────────────────┴──────────┴──────────┴──────────────────────────────┴──────

arij@vagrant:~/app-starter/app-starter$
```

2. Vérifier sous quel utilisateur tourne le conteneur

```
arij@vagrant:~/app-starter/app-starter$ docker run --rm visits-app:naive id
uid=0(root) gid=0(root) groups=0(root)
arij@vagrant:~/app-starter/app-starter$
```

Partie 3 : Dockerfile multi-stage

1. Créer le fichier .dockerignore

```
 GNU nano 7.2                                          .dockerignore *
node_modules
dist
.git
.env
Dockerfile*
compose.yaml
*.md
```

2. Compléter le Dockerfile multi-stage

```
 GNU nano 7.2                                          Dockerfile *
# ========================================
# Étape 1 : deps (installation des dépendances)
# ========================================
FROM node:24-alpine AS deps
WORKDIR /app
# On copie UNIQUEMENT les fichiers de dépendances pour maximiser le cache
COPY package*.json ./
RUN npm ci

# ========================================
# Étape 2 : build (compilation TypeScript)
# ========================================
FROM node:24-alpine AS build
WORKDIR /app
# On récupère les node_modules de l'étape deps
COPY --from=deps /app/node_modules ./node_modules
# On copie le reste des sources
COPY . .
RUN npm run build
# On supprime les dépendances de dev pour alléger l'image finale
RUN npm prune --omit=dev

# ========================================
# Étape 3 : runtime (image finale ultra-légère)
# ========================================
FROM node:24-alpine AS runtime
ENV NODE_ENV=production
WORKDIR /app
```

3. Construire l'image multi-stage

```
arij@vagrant:~/app-starter/app-starter$ time docker build -t visits-app:multistage .
[+] Building 35.8s (16/16) FINISHED                                                                                 docker:default
 => [internal] load build definition from Dockerfile                                                                          0.0s
 => => transferring dockerfile: 1.56kB                                                                                        0.0s
 => [internal] load metadata for docker.io/library/node:24-alpine                                                             2.4s
 => [internal] load .dockerignore                                                                                             0.0s
 => => transferring context: 98B                                                                                              0.0s
 => [internal] load build context                                                                                             0.2s
 => => transferring context: 289B                                                                                             0.2s
 => [deps 1/4] FROM docker.io/library/node:24-alpine@sha256:06d7adf3e0ef61a13f1d28682726366f97a15aa1888c934d3ecec86c33918b53  17.1s
 => => resolve docker.io/library/node:24-alpine@sha256:06d7adf3e0ef61a13f1d28682726366f97a15aa1888c934d3ecec86c33918b53        0.0s
 => => sha256:f0a5c9e3ce7dcc7fca3a816f70b3d42bc5324066b627223af466eff9030e171f 1.26MB / 1.26MB                                1.2s
 => => sha256:109c2f814f468711ac666fb9494ce76528e972ee8636468ad822267a0af959e5 443B / 443B                                    0.7s
 => => sha256:15577f11e05afbda845cbd2903e55f0a3da41dc2a576435999474e7a2654651b 57.37MB / 57.37MB                             14.6s
 => => extracting sha256:15577f11e05afbda845cbd2903e55f0a3da41dc2a576435999474e7a2654651b                                     2.3s
 => => extracting sha256:f0a5c9e3ce7dcc7fca3a816f70b3d42bc5324066b627223af466eff9030e171f                                     0.1s
```

4. Comparer les deux images

```
arij@vagrant:~/app-starter/app-starter$ docker image ls visits-app
                                                                                                    i Info →   U In Use
IMAGE                    ID             DISK USAGE   CONTENT SIZE   EXTRA
visits-app:multistage    b331135747787        250MB         63.3MB
visits-app:naive         55ef1630e69e        1.67GB          417MB
```

5. Vérifie l'utilisateur

```
arij@vagrant:~/app-starter/app-starter$ docker run --rm visits-app:multistage id
uid=1000(node) gid=1000(node) groups=1000(node),1000(node)
arij@vagrant:~/app-starter/app-starter$
```

Partie 4 : Pile Docker Compose

1. Compléter le fichier compose.yaml

```
 GNU nano 7.2                                                compose.yaml *
version: '3.8'

services:
  db:
    image: postgres:15-alpine
    environment:
      POSTGRES_USER: ${POSTGRES_USER}
      POSTGRES_PASSWORD: ${POSTGRES_PASSWORD}
      POSTGRES_DB: ${POSTGRES_DB}
    volumes:
      - pgdata:/var/lib/postgresql/data
    networks:
      - backend
    healthcheck:
      test: ["CMD-SHELL", "pg_isready -U ${POSTGRES_USER} -d ${POSTGRES_DB}"]
      interval: 5s
      timeout: 5s
      retries: 5

  app:
    build: .
    ports:
      - "3000:3000"
    environment:
      PGUSER: ${POSTGRES_USER}
      PGPASSWORD: ${POSTGRES_PASSWORD}
      PGDATABASE: ${POSTGRES_DB}
      PGHOST: db
    depends_on:
```

2. Valider la configuration

```
arij@vagrant:~/app-starter/app-starter$ docker compose config
WARN[0000] /home/arij/app-starter/app-starter/compose.yaml: the attribute `version` is obsolete, it will be ignored, please remove it
 to avoid potential confusion
name: app-starter
services:
  adminer:
    image: adminer:latest
    networks:
      backend: null
    ports:
      - mode: ingress
        target: 8080
        published: "8080"
        protocol: tcp
  app:
    build:
      context: /home/arij/app-starter/app-starter
      dockerfile: Dockerfile
    depends_on:
      db:
        condition: service_healthy
        required: true
    environment:
      PGDATABASE: visits
      PGHOST: db
```

3. Lancer la pile en arrière-plan

```
arij@vagrant:~/app-starter/app-starter$ docker compose up -d --build
WARN[0000] /home/arij/app-starter/app-starter/compose.yaml: the attribute `version` is obsolete, it will be ignored, please remove it
 to avoid potential confusion
[+] Running 31/31
 ✔adminer Pulled                                                                                                          25.2s
   ✔540cbae3c49d Pull complete                                                                                            10.5s
   ✔d3e42d5661e3 Pull complete                                                                                             1.3s
   ✔376608520fec Pull complete                                                                                             5.2s
   ✔7e0cb78c1bd4 Pull complete                                                                                             1.4s
   ✔2c8980f3dbe0 Pull complete                                                                                             5.3s
   ✔4f4fb700ef54 Pull complete                                                                                             1.4s
```

```
 ✔app-starter-app                Built                                                                                   0.0s
 ✔Network app-starter_backend    Created                                                                                 0.1s
 ✔Network app-starter_frontend   Created                                                                                 0.1s
 ✔Volume app-starter_pgdata      Created                                                                                 0.0s
 ✔Container app-starter-db-1     Healthy                                                                                 6.5s
 ✔Container app-starter-adminer-1 Started                                                                                1.2s
 ✔Container app-starter-app-1    Started                                                                                 7.0s
```

4.Vérifier l'état

```
arij@vagrant:~/app-starter/app-starter$ docker compose ps
WARN[0000] /home/arij/app-starter/app-starter/compose.yaml: the attribute `version` is obsolete, it will be ignored, please remove it
 to avoid potential confusion
NAME                   IMAGE                COMMAND                  SERVICE    CREATED              STATUS                        PO
RTS
app-starter-adminer-1  adminer:latest       "entrypoint.sh docke…"   adminer    About a minute ago   Up About a minute             0.
0.0.0:8080->8080/tcp, [::]:8080->8080/tcp
app-starter-app-1      app-starter-app      "docker-entrypoint.s…"   app        About a minute ago   Up About a minute (healthy)   0.
0.0.0:3000->3000/tcp, [::]:3000->3000/tcp
app-starter-db-1       postgres:15-alpine   "docker-entrypoint.s…"   db         About a minute ago   Up About a minute (healthy)   54
32/tcp
arij@vagrant:~/app-starter/app-starter$
```

5. Tester l'application

```
arij@vagrant:~/app-starter/app-starter$ curl http://localhost:3000/
{"message":"Bonjour depuis le TP7 - Version modifiée pour le cache","visits":1,"host":"bea13006193d"}arij@vagrant:~/app-starter/app-s
tarter$ curl http://localhost:3000/healtcurl http://localhost:3000/health
{"status":"ok"}arij@vagrant:~/app-starter/app-starter$
```

6. Tester



7.Relancer la pile :

```
arij@vagrant:~/app-starter/app-starter$ docker compose up -d
WARN[0000] /home/arij/app-starter/app-starter/compose.yaml: the attribute `version` is obsolete, it will be ignored, please remove it
 to avoid potential confusion
[+] Running 5/5
 ✔Network app-starter_backend      Created                                                                               0.1s
 ✔Network app-starter_frontend     Created                                                                               0.1s
 ✔Container app-starter-adminer-1  Started                                                                               0.8s
 ✔Container app-starter-db-1       Healthy                                                                               6.3s
 ✔Container app-starter-app-1      Started                                                                               6.5s
arij@vagrant:~/app-starter/app-starter$
```

8. Vérifie le compteur :

```
arij@vagrant:~/app-starter/app-starter$ curl http://localhost:3000/
{"message":"Bonjour depuis le TP7 - Version modifiée pour le cache","visits":2,"host":"6156c04fc573"}arij@vagrant:~/app-starter/app-s
tarter$
```

9. Prouver l'isolation réseau

```
arij@vagrant:~/app-starter/app-starter$ psql -h localhost -p 5432 -U visits -d visits
psql: error: connection to server at "localhost" (127.0.0.1), port 5432 failed: Connection refused
        Is the server running on that host and accepting TCP/IP connections?
arij@vagrant:~/app-starter/app-starter$
```

10.Le test de mise à l'échelle

```
arij@vagrant:~/app-starter/app-starter$ docker compose up -d --scale app=2
WARN[0000] /home/arij/app-starter/app-starter/compose.yaml: the attribute `version` is obsolete, it will be ignored, please remove it
 to avoid potential confusion
[+] Running 3/4
 ✔Container app-starter-adminer-1  Running                                                                                    0.0s
 ✔Container app-starter-db-1       Healthy                                                                                    0.6s
 ✔Container app-starter-app-1      Running                                                                                    0.0s
 ⠴ Container app-starter-app-2     Starting                                                                                   1.2s
Error response from daemon: failed to set up container networking: driver failed programming external connectivity on endpoint app-st
arter-app-2 (7f5c2591b088ab27684654b49b1521c44dc146674520bc45f0fd7800ee3c244e): Bind for :::3000 failed: port is already allocated
arij@vagrant:~/app-starter/app-starter$
```

Partie 5 : Buildx et images multi-architecture

1. Activer l'émulation ARM64

```
arij@vagrant:~/app-starter/app-starter$ docker run --privileged --rm tonistiigi/binfmt --install arm64
Unable to find image 'tonistiigi/binfmt:latest' locally
latest: Pulling from tonistiigi/binfmt
0c46aa6e9c41: Pull complete
56685a25fbf3: Pull complete
44136fa355b3: Already exists
76bf583f0cc9: Download complete
c9e92217fbff: Download complete
Digest: sha256:400a4873b838d1b89194d982c45e5fb3cda4593fbfd7e08a02e76b03b21166f0
Status: Downloaded newer image for tonistiigi/binfmt:latest
installing: arm64 OK
{
  "supported": [
    "linux/amd64",
    "linux/amd64/v2",
    "linux/amd64/v3",
    "linux/arm64",
    "linux/386"
  ],
  "emulators": [
    "python3.12",
    "qemu-aarch64"
  ]
}
```

2. Créer le builder multi-architecture

```
arij@vagrant:~/app-starter/app-starter$ docker buildx create --name multiarch --use
multiarch
arij@vagrant:~/app-starter/app-starter$ docker buildx inspect --bootstrap
[+] Building 39.8s (1/1) FINISHED
 => [internal] booting buildkit                                                                                              39.8s
 => => pulling image moby/buildkit:buildx-stable-1                                                                           37.9s
 => => creating container buildx_buildkit_multiarch0                                                                          2.0s
Name:          multiarch
Driver:        docker-container
Last Activity: 2026-10-28 22:00:34 +0000 UTC

Nodes:
Name:     multiarch0
Endpoint: unix:///var/run/docker.sock
Error:    Get "http://%2Fvar%2Frun%2Fdocker.sock/v1.51/containers/buildx_buildkit_multiarch0/json": context deadline exceeded
arij@vagrant:~/app-starter/app-starter$
```

3. Construire et pousser l'image

```
arij@vagrant:~/app-starter/app-starter$ docker buildx build --platform linux/amd64,linux/arm64 \
  -t arijbelaid/visits-app:1.0.0 --push .
[+] Building 121.9s (23/28)                                                                    docker-container:multiarch
[+] Building 281.9s (30/30) FINISHED                                                           docker-container:multiarch
 => [internal] load build definition from Dockerfile                                                              0.0s
 => => transferring dockerfile: 1.56kB                                                                            0.0s
 => [linux/arm64 internal] load metadata for docker.io/library/node:24-alpine                                     2.5s
 => [linux/amd64 internal] load metadata for docker.io/library/node:24-alpine                                     2.5s
 => [auth] library/node:pull token for registry-1.docker.io                                                       0.0s
 => [internal] load .dockerignore                                                                                 0.0s
 => => transferring context: 98B                                                                                  0.0s
 => [internal] load build context                                                                                 0.0s
 => => transferring context: 42.77kB                                                                              0.0s
 => [linux/amd64 deps 1/4] FROM docker.io/library/node:24-alpine@sha256:06d7adf3e0ef61a13f1d28682726366f97a15aa1888c934d3ecec  39.0s
 => => resolve docker.io/library/node:24-alpine@sha256:06d7adf3e0ef61a13f1d28682726366f97a15aa1888c934d3ecec86c33918b53  0.0s
 => => sha256:109c2f814f468711ac666fb9494ce76528e972ee8636468ad822267a0af959e5 443B / 443B                        1.0s
 => => sha256:f0a5c9e3ce7dcc7fca3a816f70b3d42bc5324066b627223af466eff9030e171f 1.26MB / 1.26MB                    1.6s
 => => sha256:15577f11e05afbda845cbd2903e55f0a3da41dc2a5764359994 74e7a2654651b 57.37MB / 57.37MB                32.9s
 => => sha256:7b229517ed222597277036d94892173207d01a731ab3a88eeb1de2e963a2c788 3.85MB / 3.85MB                    2.1s
 => => extracting sha256:7b229517ed222597277036d94892173207d01a731ab3a88eeb1de2e963a2c788                         0.2s
 => => extracting sha256:15577f11e05afbda845cbd2903e55f0a3da41dc2a5764359994 74e7a2654651b                       4.6s
 => => extracting sha256:f0a5c9e3ce7dcc7fca3a816f70b3d42bc5324066b627223af466eff9030e171f                         0.2s
 => => extracting sha256:109c2f814f468711ac666fb9494ce76528e972ee8636468ad822267a0af959e5                         0.0s
 => [linux/arm64 deps 1/4] FROM docker.io/library/node:24-alpine@sha256:06d7adf3e0ef61a13f1d28682726366f97a15aa1888c934d3ecec  33.8s
 => => resolve docker.io/library/node:24-alpine@sha256:06d7adf3e0ef61a13f1d28682726366f97a15aa1888c934d3ecec86c33918b53  0.0s
 => => sha256:3e52994cb01f70571f4511a3eddd348464ab581edba079f690e4d7324584ed7e 445B / 445B                        0.3s
 => => sha256:5cef607e5e751dc80266eb75d9b391b55c7c63dca5c35acf78b96b03c69b6452 1.26MB / 1.26MB                    0.8s
 => => sha256:ddcee9e51d95aa14275aa5eb7476143104fb2ff2a88767f2e4b4adeaac9b251b 56.81MB / 56.81MB                 29.8s
 => => sha256:d3531b7a4f217bdc3426c57bcff84c170f086f456a39689d4a615c6d026c4be0 4.19MB / 4.19MB                    2.2s
 => => extracting sha256:d3531b7a4f217bdc3426c57bcff84c170f086f456a39689d4a615c6d026c4be0                         0.2s
 => => extracting sha256:ddcee9e51d95aa14275aa5eb7476143104fb2ff2a88767f2e4b4adeaac9b251b                         3.8s
 => => extracting sha256:5cef607e5e751dc80266eb75d9b391b55c7c63dca5c35acf78b96b03c69b6452                         0.1s
 => => extracting sha256:3e52994cb01f70571f4511a3eddd348464ab581edba079f690e4d7324584ed7e                         0.0s
```

4. Vérifier le manifeste

```
arij@vagrant:~/app-starter/app-starter$ docker buildx imagetools inspect arijbelaid/visits-app:1.0.0
Name:      docker.io/arijbelaid/visits-app:1.0.0
MediaType: application/vnd.oci.image.index.v1+json
Digest:    sha256:4ed5cbb836167502d321c4f71d864e33319f02842935f81b33eab5f0e5c00b4b

Manifests:
  Name:        docker.io/arijbelaid/visits-app:1.0.0@sha256:6c97e9e016fb5402fd94966d09dd39f4a86e198f86263cc941286272970e2bb7
  MediaType:   application/vnd.oci.image.manifest.v1+json
  Platform:    linux/amd64

  Name:        docker.io/arijbelaid/visits-app:1.0.0@sha256:c744c9a55bdaf0571aae9410737bfd3a74fcf3305097e7cf2239957ed15c5bf8
  MediaType:   application/vnd.oci.image.manifest.v1+json
  Platform:    linux/arm64

  Name:        docker.io/arijbelaid/visits-app:1.0.0@sha256:f6e11264f7a1718afc1cdc40a7545ae0e91f7d04502678c353b56a5ad574098c
  MediaType:   application/vnd.oci.image.manifest.v1+json
  Platform:    unknown/unknown
  Annotations:
    vnd.docker.reference.digest: sha256:6c97e9e016fb5402fd94966d09dd39f4a86e198f86263cc941286272970e2bb7
    vnd.docker.reference.type:   attestation-manifest

  Name:        docker.io/arijbelaid/visits-app:1.0.0@sha256:baf31d0949913894bb18485b6032d9d641ac5de13548fe252a0579a598bd7433
  MediaType:   application/vnd.oci.image.manifest.v1+json
  Platform:    unknown/unknown
  Annotations:
    vnd.docker.reference.digest: sha256:c744c9a55bdaf0571aae9410737bfd3a74fcf3305097e7cf2239957ed15c5bf8
    vnd.docker.reference.type:   attestation-manifest
```